Basics of automation technology

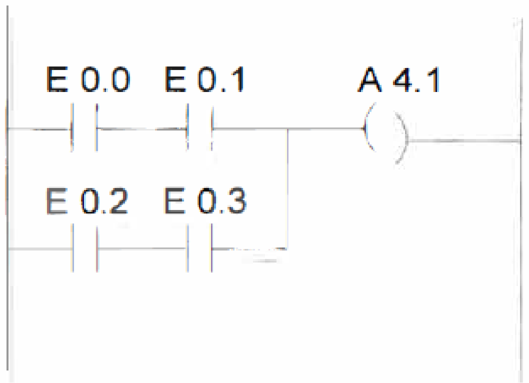
Samppa Alanen

Ref. Seppo Rantapuska

Jyväskylän ammattikorkeakoulu
University of Applied Sciences

Presentation of the control program

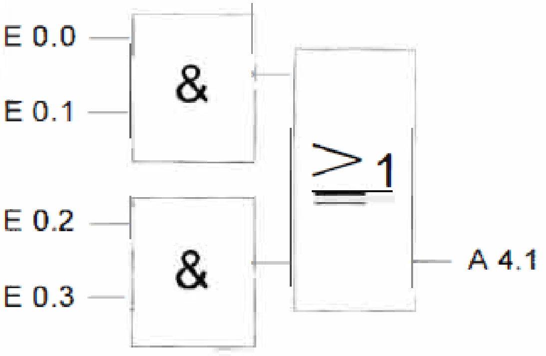
Ladder LAD



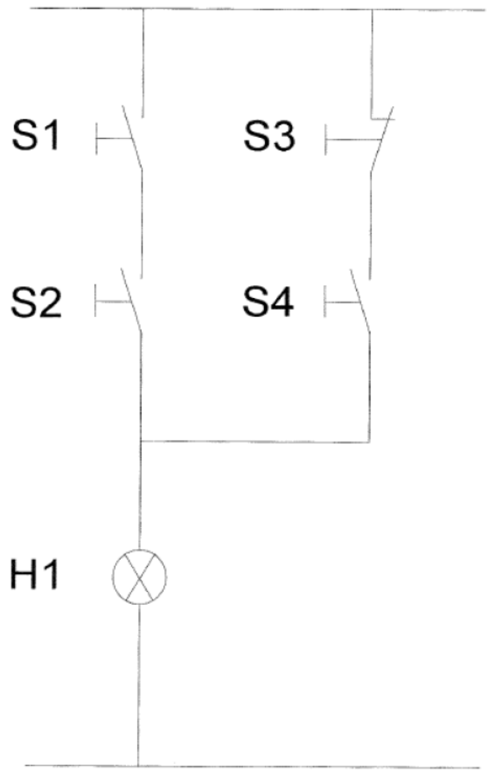
STL

```
U    E 0.0
U    E 0.1
O
U    E 0.2
U    E 0.3
=    A 4.1
```

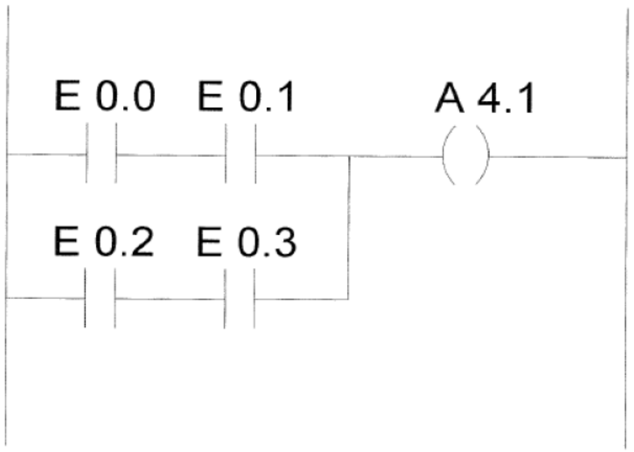
Function blocks FBD



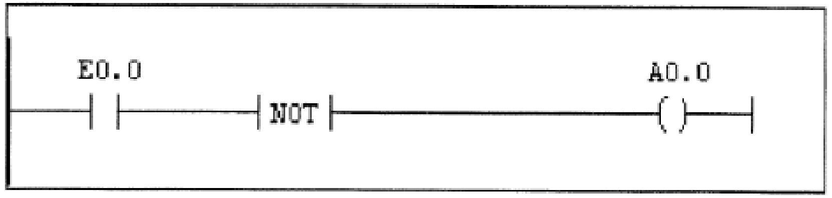
Circuit diagram



Contact diagram LAD

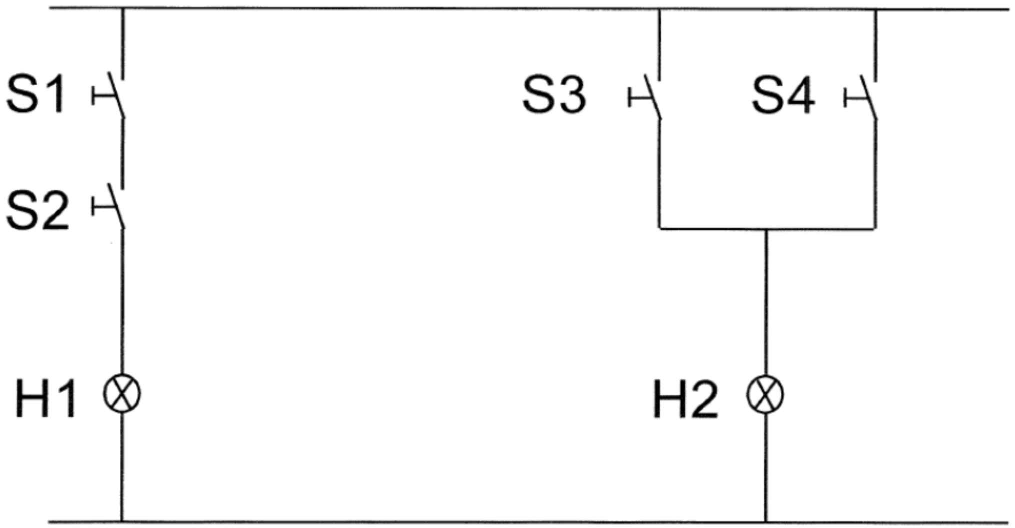


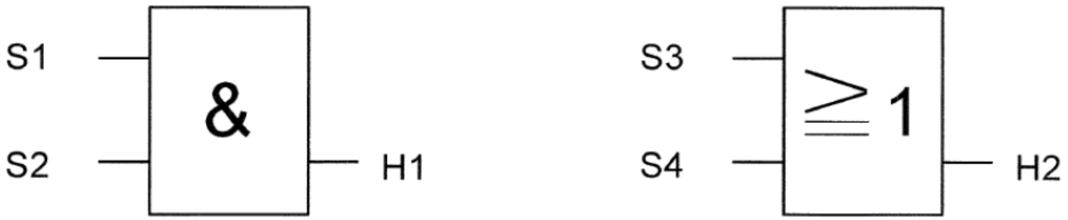
- Contact diagram (ger. KOP, eng. LAD) resembles a circuit diagram. On the screen, the circuits are not vertical, but horizontal.
- The symbols --] [-- and --] / [-- do not represent switches, but a logical query of signal states "1" and "0".
- So it is with outputs. The symbols --( )-- and ---(/)-- describe a decision operation or decision inverted.
- The inversion is presented in the SIMATIC S7 program using the NOT command:



jamk

Binary operations



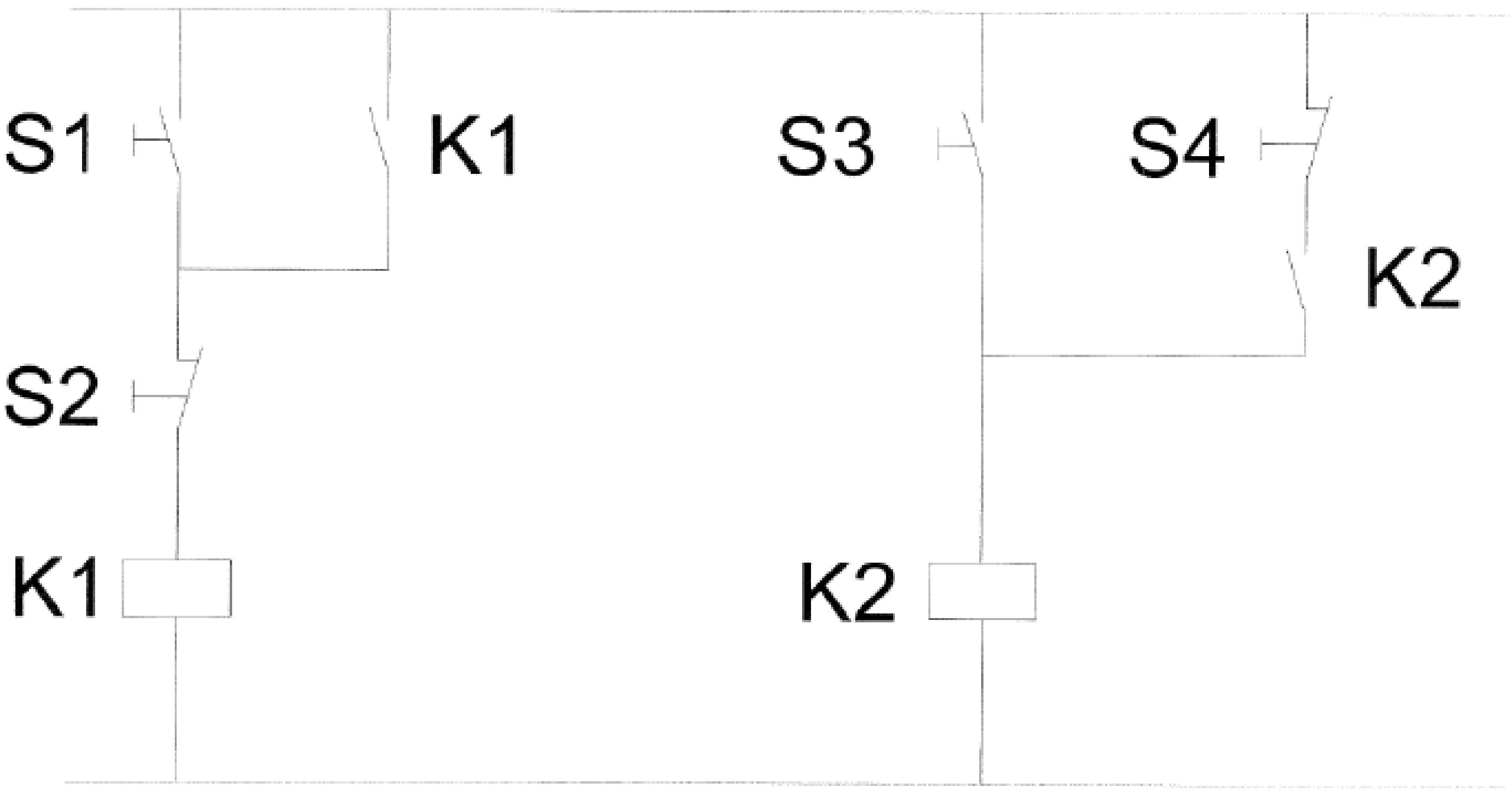


AND OR

- Lamp H1 is on only when S1 and S2 are closed. In this case, a connection is established between inputs and outputs. This is called AND locking.
- Switches S1 and S2 should be turned on so that the lamp H1 is lit.
- The symbol for this lock is &.
- For the lamp H2 to be lit must S3 or S4 be closed. In this case, a connection is established between inputs and outputs. This is called OR locking.
- Switches S3 or S4 must be on so that the lamp H2 is lit.
- The symbol for this lock is ≥ 1.

- In the automation system, the program asks for input status in the following way:
- If there is a voltage (most often about 13-30VDC), then the input has a signal mode "1".
- If there is no voltage (most often <5VDC), then the input has a signal state "0".
- With signal sensors, it should be noted that the automation system does not have the ability to know whether the sensor is implemented with an opening or closing contact.
- Logic can ask for both "1", and "0" status of the input.

Memory coil



Close/reset has high priority
SR-flip flop

Open/set has high priority
SR-flip flop